

Lecture 9: Reservoir Computing
Echo State Networks, Cellular Automata, Qubits

Deep Learning in Classical and Quantum Physics

Marcin Płodzień

Atomic Optics Department
Institute of Theoretical Physics
Jagiellonian University in Kraków

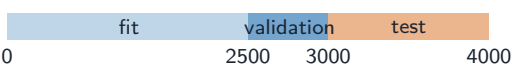
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Motivation



Recap of the Forecasting Task

record	signal s_k
Lorenz	x of Lorenz-63 convection, $\Delta t = 0.02$
MG 15, 25, 35	Mackey–Glass delay equation, delay τ_{MG}, unit step; MG 15 periodic
Santa Fe	measured intensity of a far-infrared laser



4000 samples per record, scaled to $[0, 1]$ on the training block

Forecast and score

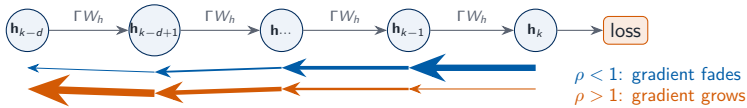
$\tilde{\mathbf{y}}_k = (\tilde{s}_{k+1}, \ldots, \tilde{s}_{k+50})$ from $s_k, s_{k-1}, \ldots$

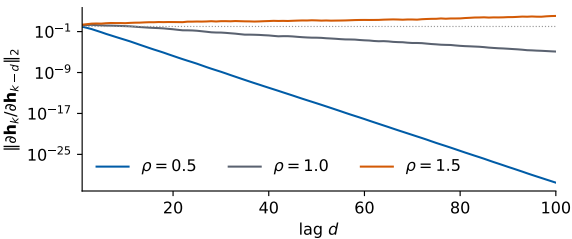
$C(\tau) := \mathrm{corr}^2\,(\tilde{s}_{k+\tau}, s_{k+\tau})$ over the test origins k

$C_{\mathrm{tot}} := \sum_{\tau=1}^{50} C(\tau) \leq 50$

- persistence $\tilde{s}_{k+\tau} = s_k$; AR: linear in $s_k, \ldots, s_{k-p+1}$; NVAR: plus products $s_{k-i}s_{k-j}$
- linear readout by ridge regression, one solve for all horizons

Gradients through Time Vanish or Explode





largest singular value of $\partial \mathbf{h}_k / \partial \mathbf{h}_{k-d}$ against the lag d, random tanh network

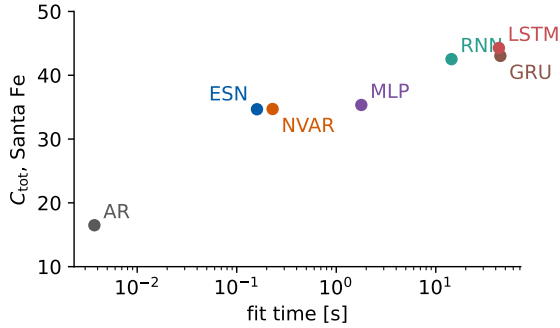
Product of step Jacobians

$$\frac{\partial \mathbf{h}_k}{\partial \mathbf{h}_{k-d}} = \Gamma_k W_h \cdots \Gamma_{k-d+1} W_h$$

$\Gamma_k := \mathrm{diag}(1 - \mathbf{h}_k^2)$; ρ: spectral radius of W_h

- an input d steps old enters through d factors

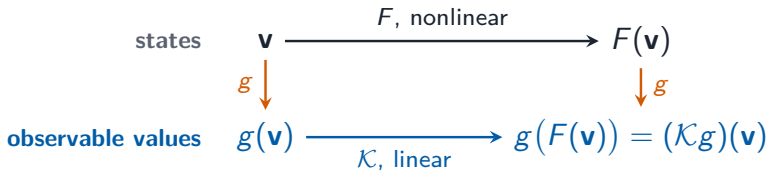
Trainable Recurrence Has a Price



forecast quality C_{tot} against fit time; trained networks and a reservoir (ESN)

- fixing the gradients (gated units, LSTM and GRU) makes training possible, but every weight is still fitted by gradient descent through the whole history
- a reservoir with one linear fit costs a fraction of the time

Koopman: Nonlinear Dynamics Is Linear on Observables

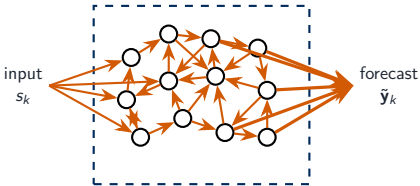


- Koopman operator $\mathcal{K}g := g \circ F$: linear in g, even for nonlinear F
- with enough observables, forecasting is a linear map
- a reservoir supplies the observables: a fixed embedding of the input history

B. O. Koopman, Proc. Natl. Acad. Sci. USA 17, 315 (1931)

Fixed Embedding, Trained Linear Readout

recurrent network: all weights trained

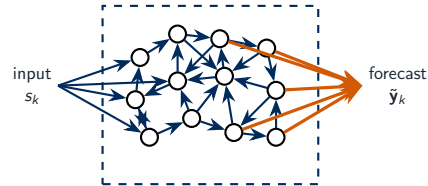


$$\mathbf{h}_k = \tanh\left(W_h\,\mathbf{h}_{k-1} + W_x\,s_k\right)$$

$$\tilde{\mathbf{y}}_k = W_{\text{out}}^{\top}\,\mathbf{h}_k + \mathbf{b}$$

gradient descent through the whole history

echo state network: only the readout trained



$$\mathbf{r}_k = \tanh\left(W\,\mathbf{r}_{k-1} + \mathbf{w}_{\text{in}}\,s_k\right)$$

$$\tilde{\mathbf{y}}_k = W_{\text{out}}^{\top}\,\mathbf{r}_k + \mathbf{b}$$

$W, \mathbf{w}_{\text{in}}$ random and fixed; $W_{\text{out}}, \mathbf{b}$ by one ridge solve

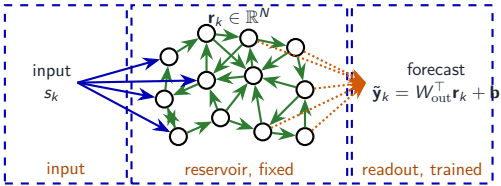
trained random, fixed

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Reservoir Computing

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Echo State Network: Fixed Reservoir, Linear Readout



Idea

$(s_1, \ldots, s_k) \xrightarrow{\text{fixed}} \mathbf{r}_k \xrightarrow{\text{linear}} \tilde{\mathbf{y}}_k$

train only $W_{\text{out}}, \mathbf{b}$, one ridge solve; $\mathbf{r}_k$ feeds step $k+1$ and the forecast

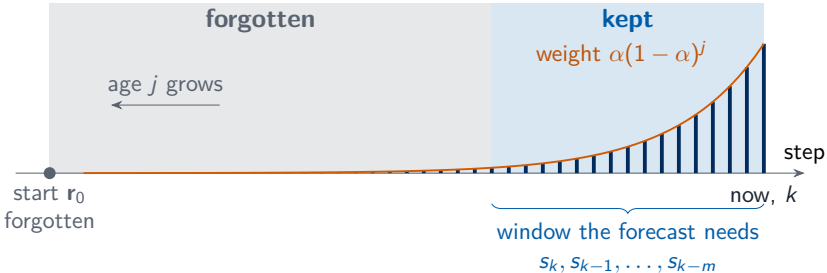
Leaky state update and linear readout

$$\mathbf{r}_k = (1-\alpha)\,\mathbf{r}_{k-1} + \alpha \tanh\left(W \mathbf{r}_{k-1} + \mathbf{w}_{\text{in}} s_k\right), \qquad \tilde{\mathbf{y}}_k = W_{\text{out}}^\top \mathbf{r}_k + \mathbf{b}$$

- W: sparse random $N \times N$ (10% Gaussian entries), rescaled so that $\max_i |\lambda_i| = \rho$, the **spectral radius** (complex λ_i, any square matrix)
- ρ: feedback strength; $\rho < 1$ forgets, $\rho \approx 1$ longest echoes, $\rho > 1$ may amplify
- $\mathbf{w}_{\text{in}}$: random in $[-\sigma_{\text{in}}, \sigma_{\text{in}}]$, **input scale** σ_{in}; **leak rate** $\alpha \in (0, 1]$

H. Jaeger, GMD Report 148 (2001)

A Reservoir Must Keep the Recent Past and Forget the Rest



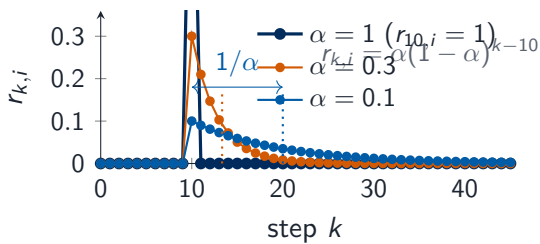
- remember: $s_{k+\tau}$ depends on the window $s_k, \ldots, s_{k-m}$, and the readout sees only $\mathbf{r}_k$
- forget: the start $\mathbf{r}_0$ and inputs older than the window
- balance: the memory window should match the window the forecast needs; the leak rate α and the spectral radius ρ set it

Why Forget, and How Fast: the Leak Rate

Three reasons to forget (1) the start $\mathbf{r}_0$ is arbitrary (2) old inputs say little after a few $T_\lambda = 1/\lambda_1$ (Lorenz: about 55 samples) but fill the N components (3) for $\rho > 1$ differences and noise grow

The leak rate sets how long one input is remembered

$$r_{k,i} = (1-\alpha)\, r_{k-1,i} + \alpha\, g_{k,i}, \qquad g_{k,i} := \left[\tanh\left(W\mathbf{r}_{k-1} + \mathbf{w}_{\text{in}} s_k\right)\right]_i$$

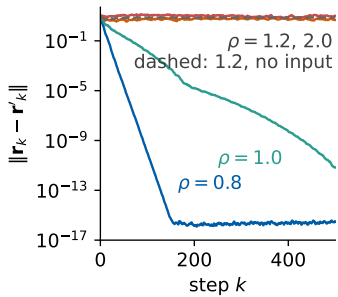


one input at $k = 10$ ($g_{10,i} = 1$, all other $g_{k,i} = 0$); an input j steps old weighs $\alpha(1-\alpha)^j$: memory about $1/\alpha$ steps (RC filter, $\alpha = \Delta t/\tau$); ρ sets the echoes among units

- why a leak: a slowly varying, finely sampled signal changes little per step; $\alpha < 1$ slows the units to its time scale

The State Forgets Its Start and Old Inputs

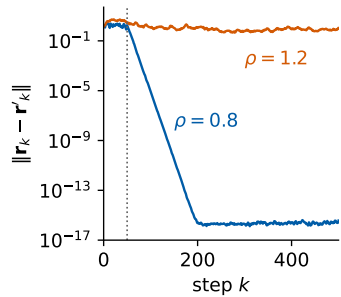
echo state property



same input, $\mathbf{r}_0 \neq \mathbf{r}'_0$: $\|\mathbf{r}_k - \mathbf{r}'_k\| \to 0$

forget the start

fading memory



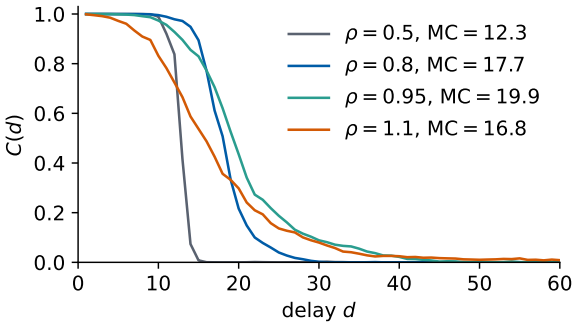
inputs equal after $k = 50$: $\|\mathbf{r}_k - \mathbf{r}'_k\| \to 0$

forget old inputs

$N = 100$, $\alpha = 1$, $\sigma_{\text{in}} = 0.5$, Santa Fe input

H. Jaeger, GMD Report 148 (2001); S. Boyd, L. O. Chua, IEEE Trans. Circuits Syst. 32, 1150 (1985)

Memory Capacity



$\alpha = 1$, $\sigma_{\text{in}} = 0.1$

J. Dambre *et al.*, Sci. Rep. 2, 514 (2012)

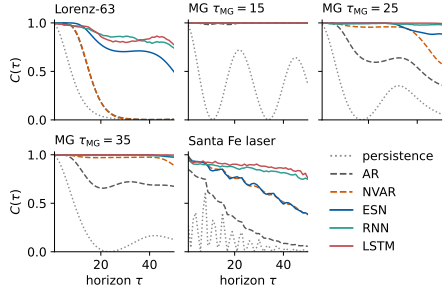
Recall of a random input

$$C(d) := \text{corr}^2(\tilde{u}_{k-d}, u_{k-d})$$
$$\text{MC} := \sum_{d=1}^{100} C(d) \leq N$$

u_k: independent random input; one readout per delay d; MC: past inputs the state still holds

- the trade-off in numbers: MC grows towards $\rho \approx 1$ and drops beyond
- MC $= 12.3$ at $\rho = 0.5$, 19.9 at 0.95, 16.8 at 1.1; bound $N = 50$

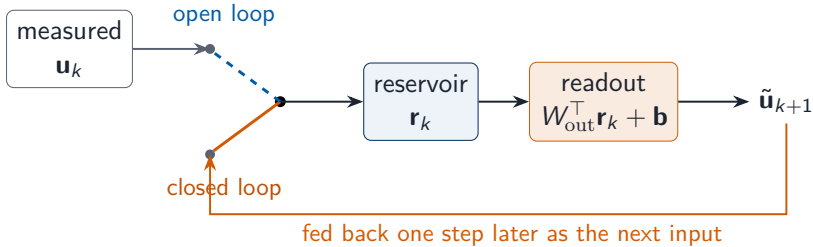
ESN Forecasts of the Five Records



$C(\tau)$: squared correlation of forecast $\tilde{s}_{k+\tau}$ and truth on the test block; persistence, AR, NVAR, ESN ($N = 300$), RNN and LSTM trained on the same data

- the ESN beats the fixed delay features most on Lorenz, where the forecast needs a long nonlinear memory
- fully trained RNN and LSTM lead on Lorenz and the laser, at the price of many passes of gradient descent

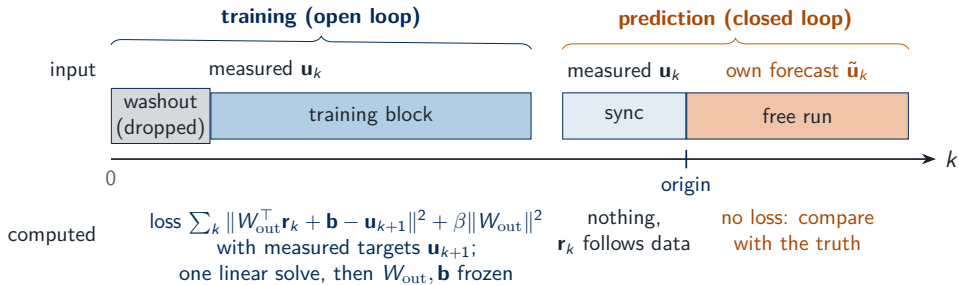
Closed Loop: the Forecast Becomes the Next Input



- closed loop: $\tilde{\mathbf{u}}_{k+1}$ replaces the measurement; an autonomous system
- use 1: forecasts beyond the data
- use 2: a model learned from the record alone; a long free run gives its attractor and λ_1, right only if the dynamics was learned

$\mathbf{u}_k$: (x, y, z) of Lorenz-63, normalised; ESN with $N = 300$; J. Pathak *et al.*, Chaos 27, 121102 (2017)

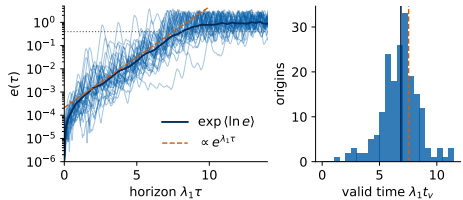
Train on Data, Predict on Its Own Forecast



- **training**: measured $\mathbf{u}_k$ in, readout fitted to $\mathbf{u}_{k+1}$
- **sync**: measured data bring $\mathbf{r}_k$ to the current state
- **free run**: $\tilde{\mathbf{u}}_{k+1}$ is the next input; error gives the valid time, the long run the climate

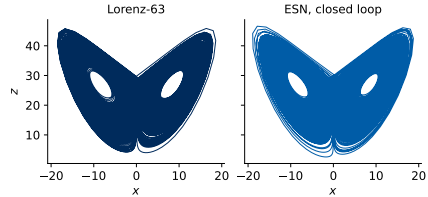
A Free Run Loses the Weather and Keeps the Climate

weather: one trajectory



left: distance $e(\tau)$ of free run and truth from one start, horizon in $T_\lambda = 1/\lambda_1$; right: valid time t_v, first $e > 0.4$ (dotted), 190 starts

climate: the attractor



(x, z) of a long true run (left) and of the free run, about 1988 T_λ (right)

- weather: errors grow as $e^{\lambda_1 \tau}$; median valid time 6.9 T_λ, as for any chaotic forecast
- climate: same attractor; largest Lyapunov exponent 0.909 against 0.904 for Lorenz-63

Requirements for a Good Reservoir

1. **echo state property**: the state forgets its start, depends only on the inputs
2. **fading memory**: recent inputs matter more than remote ones
3. **separability**: different recent histories give different states
4. **nonlinearity**: products of inputs at different delays
5. **richness**: many independent readable quantities
6. **time scales**: memory of the reservoir matches the task

- nothing here asks for a neural network: any driven system with these properties can serve

L. Grigoryeva, J.-P. Ortega, Neural Netw. 108, 495 (2018); G. Tanaka *et al.*, Neural Netw. 115, 100 (2019)

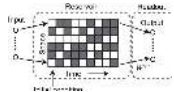
Physical Reservoirs

nothing inside is trained: any system with a nonlinear response, fading memory and many readable quantities, also interacting qubits; review: G. Tanaka *et al.*, Neural Netw. 115, 100 (2019)



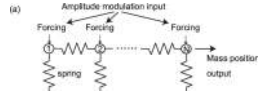
water in a bucket: the input drives waves, a camera image of the surface gives the features

C. Fernando, S. Sojakka, ECAL 2003, 588



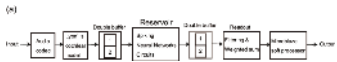
cellular automaton: input written into cells, a fixed local rule evolves them, the pattern is read out

O. Yilmaz, arXiv:1410.0162 (2014)



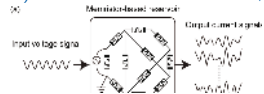
coupled oscillators: masses on springs driven by the input, positions read out

H. Hauser *et al.*, Biol. Cybern. 105, 355 (2011)



electronic circuit: one nonlinear node with a delay line, delayed samples act as virtual nodes

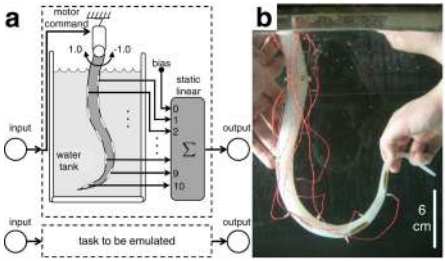
L. Appeltant *et al.*, Nat. Commun. 2, 468 (2011)



memristor network: resistance depends on the past current, so the network remembers

C. Du *et al.*, Nat. Commun. 8, 2204 (2017)

A Soft Silicone Arm as a Reservoir



(a) motor command in, 10 bend sensors out, weighted sum; (b) the arm (45 cm) in water



click: video

Input: motor angle of the base, $m(k) \in [-1, 1]$ ($\pm 46°$), one step $\approx 0.03\,\mathrm{s}$

$$m(k) = \sin\left(2\pi f_1 \frac{k}{T}\right) \sin\left(2\pi f_2 \frac{k}{T}\right) \sin\left(2\pi f_3 \frac{k}{T}\right)$$

$(f_1, f_2, f_3) = (2.11, 3.73, 4.33)$; T: time scale of the command, larger T is a slower motion

- **state**: 10 bend sensors, $x_i(k) = F_i\big(m(k), m(k-1), \dots\big)$; water drag gives nonlinearity, damped relaxation fading memory
- **readout**: $\tilde{y}_k = \sum_i w_i x_i(k) + w_0$, only w fitted
- **task memory**: NARMA-n targets depend nonlinearly on the last n commands; one body, five readouts for $n = 2, 5, 10, 15, 20$

K. Nakajima, H. Hauser, T. Li, R. Pfeifer, Sci. Rep. 5, 10487 (2015); figure from this reference

3

Cellular Automata



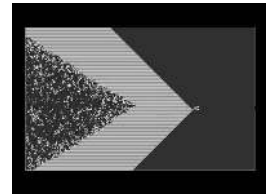
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Simple Rules Generate Complicated Patterns

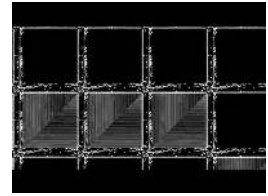
Conway's Game of Life: a grid of cells, alive or dead; the whole rule fits in one line

> born with 3 live neighbours, survives with 2 or 3

- yet: still lifes, oscillators, gliders that travel, guns that emit gliders
- patterns built from gliders compute; Life can even simulate Life itself
- complexity is not in the rule but in the dynamics it generates



epic Game of Life (Rational Animations)



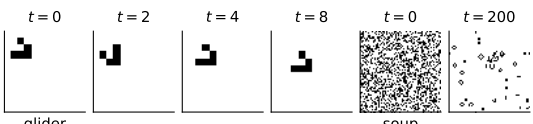
Life in life (P. Bradbury)

M. Gardner, Sci. Am. 223, 120 (1970); E. R. Berlekamp, J. H. Conway, R. K. Guy, Winning Ways for Your Mathematical Plays, Vol. 2 (Academic Press, 1982)

A Cellular Automaton: Lattice, States, One Local Rule

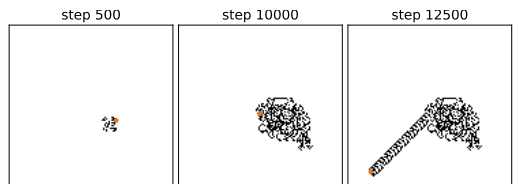
Cellular automaton

a lattice of cells; each cell in one of a **finite set of states**; all cells updated at once by **one local rule**, a finite table from the states of a cell and its neighbours to its new state



Game of Life: 2D, states $\{0, 1\}$, 8 neighbours; birth on 3, survival on 2 or 3; a glider moves, a random soup settles

Conway on inventing Life (Numberphile)



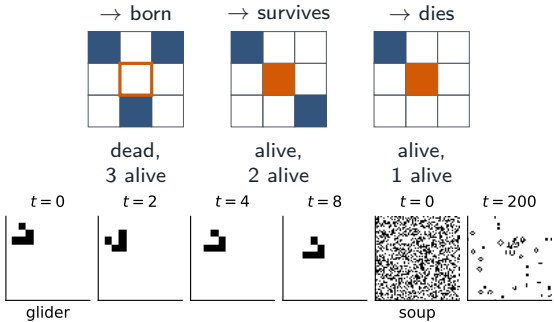
Langton's ant: on white turn right, on black turn left, flip the cell, step; chaos, then a periodic highway

Langton's ant (discovermaths)

M. Gardner, Sci. Am. 223, 120 (1970); C. G. Langton, Physica D 22, 120 (1986)

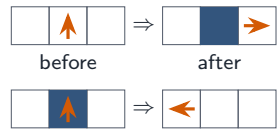
Two Rules: Game of Life and Langton's Ant

Game of Life: each cell looks at its 8 neighbours



evolved: a glider moves; a random soup settles

Langton's ant: one moving cell, the ant



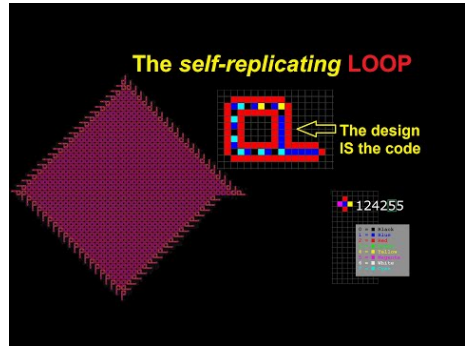
then flip the colour of the cell and step forward



evolved: about 10^4 chaotic steps, then a repeating highway

Cellular Automata Were Invented to Model Self-Reproduction

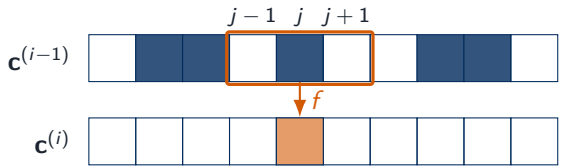
- von Neumann and Ulam: a machine that builds a copy of itself, as a grid of cells with 29 states and one local rule
- Langton's loops: 8 states; a loop carries its own description and grows copies
- simple local rules give self-reproduction, computation and complex patterns



Langton's loops: the cellular automaton that copies itself

J. von Neumann, A. W. Burks (ed.), Theory of Self-Reproducing Automata (Univ. Illinois Press, 1966);
C. G. Langton, Physica D 10, 135 (1984)

Elementary Cellular Automata (Wolfram)



ring of L cells:
cell $L-1$ next to cell 0

Elementary cellular automaton

$c_j^{(i)} \in \{0,1\}$: state of cell $j = 0, \ldots, L-1$ after i iterations

$c_j^{(i)} = f\left(c_{j-1}^{(i-1)}, c_j^{(i-1)}, c_{j+1}^{(i-1)}\right)$ for all j at once, one local rule f

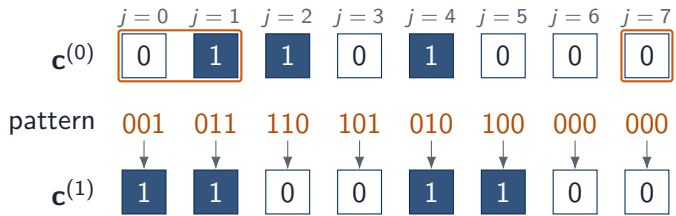
global map $\mathbf{c}^{(i)} = F\left(\mathbf{c}^{(i-1)}\right)$: discrete in space, time and states

S. Wolfram, Rev. Mod. Phys. 55, 601 (1983); S. Wolfram, A New Kind of Science (Wolfram Media, 2002);

S. Wolfram on cellular automata and rule 30 (Lex Fridman podcast clip)

One Update of Rule 30

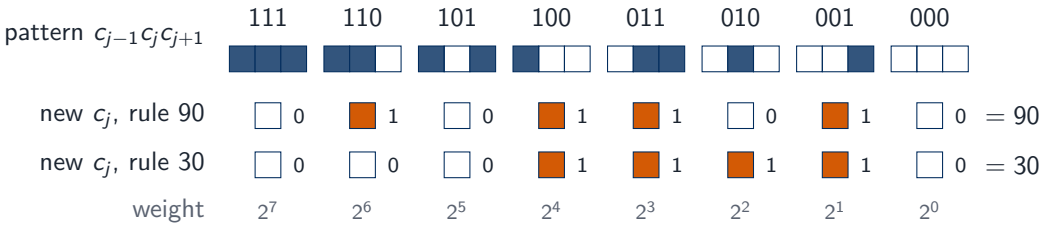
pattern $c_{j-1}c_jc_{j+1}$	111	110	101	100	011	010	001	000
new state c_j	0	0	0	1	1	1	1	0



cell 0 reads cell 7, itself and cell 1

- each cell reads its pattern $c_{j-1}c_jc_{j+1}$; the table gives its new state
- ring closed: cell 0 reads 001 and becomes 1
- all cells at once: $\mathbf{c}^{(0)} = 01101000 \;\rightarrow\; \mathbf{c}^{(1)} = 11001100$

Rule Number

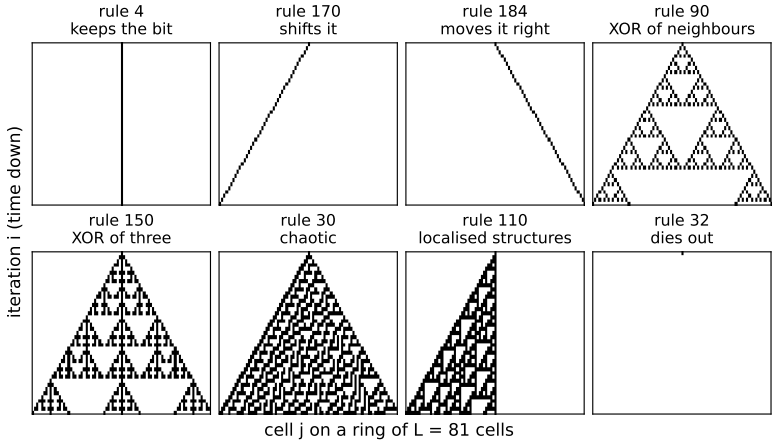


the 8 patterns of a cell and its neighbours (dark: 1); new centre state (orange: 1); weight of each column; XOR $\oplus$: 1 if the bits differ, OR $\vee$

a	b	$a \oplus b$	$a \vee b$
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	1

- rule 90: $c_j \leftarrow c_{j-1} \oplus c_{j+1}$, e.g. 110: $1 \oplus 0 = 1$
- rule 30: $c_j \leftarrow c_{j-1} \oplus (c_j \vee c_{j+1})$, e.g. 011: $0 \oplus 1 = 1$
- new states as a binary number: $00011110_2 = 30$; $2^8 = 256$ rules

One Bit, Eight Rules



- same start, a single 1 in L cells: kept, moved, copied into patterns, scrambled, erased

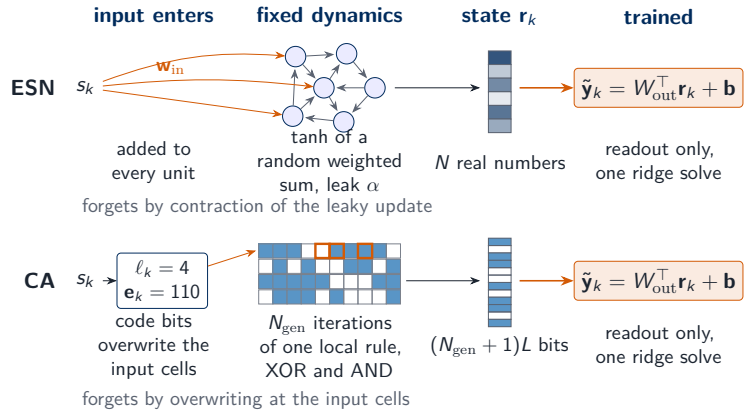
4

Cellular Automaton Reservoir



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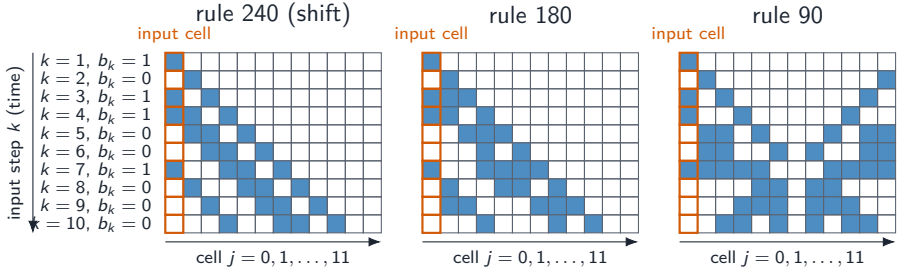
From the ESN to an Automaton



CA: cellular automaton reservoir, a ring of L bits; same ridge fit, bits read as 0 and 1

O. Yilmaz, arXiv:1410.0162 (2014)

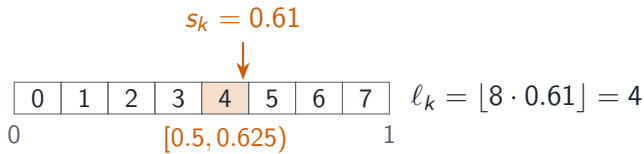
A Driven Ring Records Its Input



empty start; at step k the bit b_k replaces cell 0 (orange), then one iteration; row k: ring right after writing b_k; blue: 1

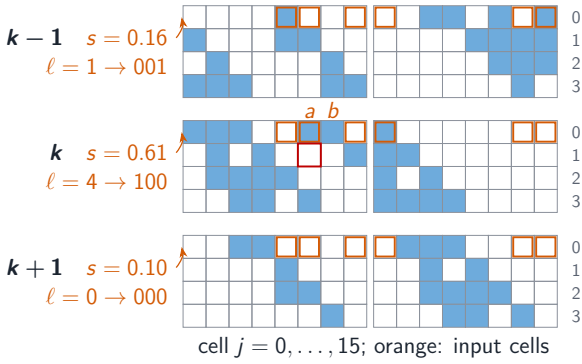
- rule 240, $c_j \leftarrow c_{j-1}$: cell d holds b_{k-d}, a delay line of the last 12 inputs
- rule 180, $c_j \leftarrow c_{j-1} \oplus c_j \oplus c_j c_{j+1}$: moves bits and multiplies neighbours
- rule 90: copies of different inputs meet and add modulo 2, unreadable

Quantisation Cuts the Signal into Q Levels



- $Q :=$ number of levels: $[0, 1]$ cut into Q bins of width $1/Q$
- level $\ell_k := \lfloor Q s_k \rfloor$, clipped to $0, \ldots, Q-1$; here $Q = 8$, forecasts $Q = 16$
- code word $\mathbf{e}_k := \ell_k$ in binary, $B = \log_2 Q$ bits: $4 \to 100$

One Input Step: Encode, Evolve, Read Out



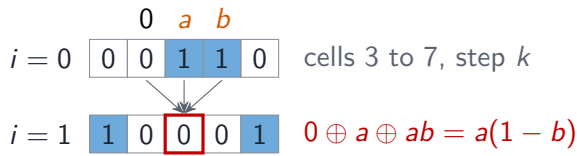
one block per input step $k-1, k, k+1$; rows: generations $i = 0, \ldots, N_{\text{gen}}$ (time down); columns: 16 cells in $R = 2$ compartments of 8; blue: 1; orange frames: input cells; a, b: the two cells that set the red cell

1. encode: $s_k = 0.61 \to \ell_k = \lfloor 8 \cdot 0.61 \rfloor = 4 \to 100$; the 3 bits overwrite the orange input cells, all other cells keep their state from step $k-1$, the memory

2. evolve: rule 180 applied $N_{\text{gen}} = 3$ times, $c_j \leftarrow c_{j-1} \oplus c_j \oplus c_j c_{j+1}$; new bits mix with old ones, the product $c_j c_{j+1}$ is the nonlinearity

3. read out: all bits of the $N_{\text{gen}} + 1$ rows of step k form $\mathbf{r}_k$ ($16 \times 4 = 64$ features), $\tilde{\mathbf{y}}_k = W_{\text{out}}^{\top} \mathbf{r}_k + \mathbf{b}$

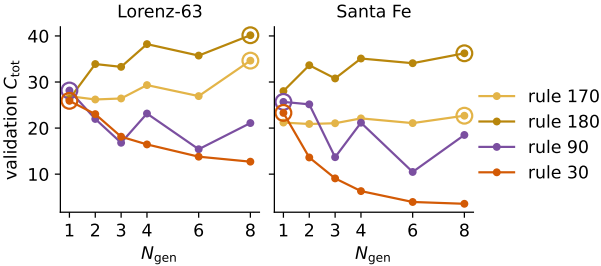
The Nonlinearity Comes from Logic on Bits



cells 3 to 7 of step k; cell 5 receives the bit a of s_k, its right neighbour holds the bit b left by s_{k-1}, its left neighbour is 0; red: one generation later

- rule 180: $c_j \leftarrow c_{j-1} \oplus c_j \oplus c_j\, c_{j+1}$; cell 5 with $c_4 = 0$, $c_5 = a$, $c_6 = b$: $0 \oplus a \oplus ab$
- $ab = a$ AND b, the product of the two bits: 1 only if both are 1
- $b = 0$: $a \oplus 0 = a$; $b = 1$: $a \oplus a = 0$; together $a(1 - b)$, here $a = b = 1 \rightarrow 0$
- $a(1 - b)$ is no weighted sum of a and b: the readout receives a product of s_k and s_{k-1} bits, like $s_k s_{k-1}$ in NVAR

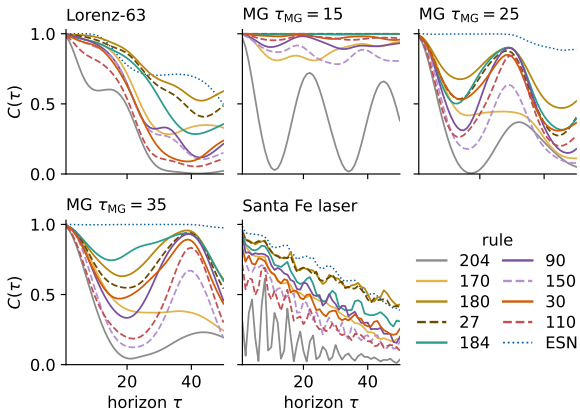
N_{gen} Trades Spacing against Length of the Record



validation C_{tot}; open circle: selected N_{gen}; $R = L_c = 16$

- chosen with β on validation, $N_{\mathrm{gen}} \in \{1, 2, 3, 4, 6, 8\}$
- 180: inputs N_{gen} cells apart, about L_c / N_{gen} per compartment; selects 8 (Lorenz), 4 (Santa Fe)
- 90, 30: each generation scrambles; select 1

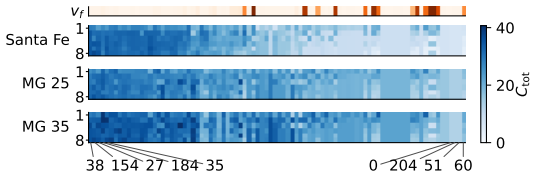
Capacity of Nine Rules against the Horizon



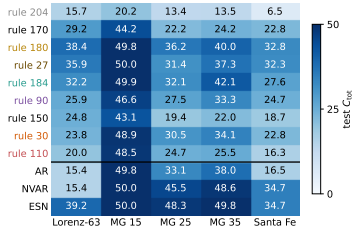
test $C(\tau)$ against the horizon τ; each rule at its selected N_{gen}; dotted: ESN; 170: $c_j \leftarrow c_{j+1}$, pure shift; 27: $1 \oplus c_j \oplus c_j c_{j+1} \oplus c_{j-1} c_{j+1}$; 184 (traffic): $c_{j-1} \oplus c_{j-1} c_j \oplus c_j c_{j+1}$

- 204 (identity): near persistence
- 180, 27, 184 (move, multiply): highest, near the ESN
- 90, 150, 30, 110 (spread): lower

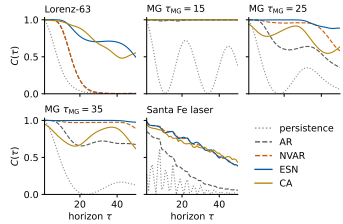
Which Automata Forecast Best



validation C_{tot}, all rules, best left; rows N_{gen}



test C_{tot} of nine rules; below the line AR, NVAR, ESN



test $C(\tau)$, selected automaton and ESN

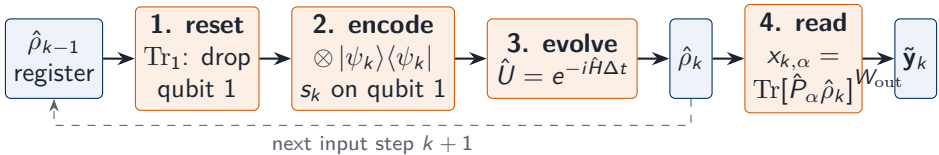
- best: one class of moving rules, 154, 166, 180, 210 (mirror images, $0 \leftrightarrow 1$); worst: 0, 51, 204
- selected 180: above the ESN on Lorenz, below on MG 25, 35
- two copies from different starts merge under 180, 210, never under 90, 30, 4

5
Quantum Reservoir



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One Step of a Quantum Reservoir



Input step k: reset, encode, evolve

$$\hat{\rho}_k = \hat{U}\big[\,|\psi_k\rangle\langle\psi_k| \otimes \mathrm{Tr}_1\,\hat{\rho}_{k-1}\big]\,\hat{U}^\dagger, \qquad \hat{U} = e^{-i\hat{H}\Delta t}$$

$$|\psi_k\rangle = \hat{R}_y(\theta_k)|0\rangle = \sqrt{1-s_k}\,|0\rangle + \sqrt{s_k}\,|1\rangle, \qquad \theta_k = 2\arcsin\sqrt{s_k}$$

Tr_1: partial trace over qubit 1; it discards the old state of qubit 1 and keeps qubits $2, \ldots, N$, the memory

Features Are Expectation Values of Pauli Strings

Pauli matrices and Pauli strings

$\hat{X} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\hat{Y} = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$, $\hat{Z} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, identity $\hat{I}$

Pauli string: one of $\hat{I}, \hat{X}, \hat{Y}, \hat{Z}$ on every qubit, e.g. $\hat{Z}_1 = \hat{Z} \otimes \hat{I} \otimes \cdots \otimes \hat{I}$, $\hat{Z}_1\hat{Z}_2 = \hat{Z} \otimes \hat{Z} \otimes \hat{I} \otimes \ldots$

Feature vector and readout

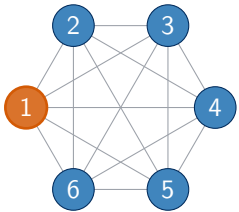
$$\mathbf{x}_k = \left(\mathrm{Tr}[\hat{P}_\alpha\, \hat{\rho}_k]\right)_{\alpha=1}^{M}, \qquad \hat{P}_\alpha \in \{\hat{X}_i, \hat{Y}_i, \hat{Z}_i, \hat{X}_i\hat{X}_j, \hat{Y}_i\hat{Y}_j, \hat{Z}_i\hat{Z}_j\},$$

$$\tilde{\mathbf{y}}_k = W_{\mathrm{out}}^{\top}\mathbf{x}_k + \mathbf{b}, \text{ only } W_{\mathrm{out}}, \mathbf{b} \text{ trained}$$

- $\mathrm{Tr}[\hat{P}\hat{\rho}]$: average of the outcomes ± 1 when $\hat{P}$ is measured (Born rule)
- $M = 3N + 3N(N-1)/2 = 63$ features for $N = 6$, out of $4^N - 1 = 4095$ numbers that fix $\hat{\rho}_k$

M. A. Nielsen, I. L. Chuang, Quantum Computation and Quantum Information (Cambridge, 2010)

Quantum Hamiltonian of the Reservoir



15 pairs, each its own J_{ij}

$N = 6$ qubits, input on qubit 1; J_s: unit of energy

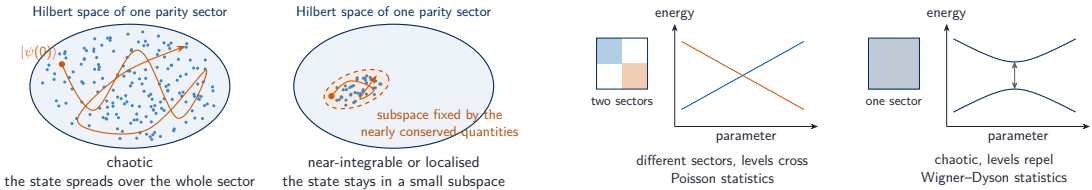
Hamiltonian

$$\hat{H} = \sum_{i<j} J_{ij}\,\hat{X}_i\hat{X}_j + \tfrac{1}{2}\sum_{i=1}^{N}(h + D_i)\,\hat{Z}_i$$

- $\hat{X}_i\hat{X}_j$ flips qubits i, j; J_{ij} uniform in $[-J_s/2, J_s/2]$
- h: the same splitting on every qubit
- D_i uniform in $[-D, D]$: a flip-flop $|01\rangle \to |10\rangle$ on i, j costs $|D_i - D_j|$

K. Fujii, K. Nakajima, PR Applied 8, 024030 (2017); R. Martínez-Peña *et al.*, PRL 127, 100502 (2021)

Chaos Spreads the State, Conservation Confines It

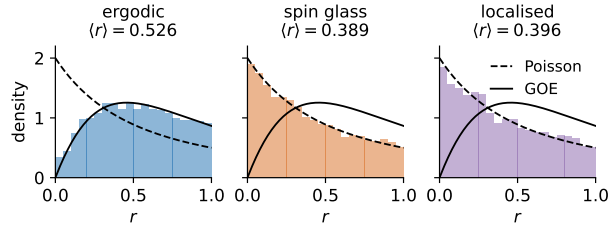


- chaotic: the state explores the whole sector, the input spreads into many-qubit correlations (scrambling), the start is forgotten; near-integrable or localised: conserved quantities confine the state
- energies $\hat{H}|E_n\rangle = E_n|E_n\rangle$; different sectors: no matrix element of $\hat{H}$ between them, levels cross; one sector: a coupling V keeps levels $\geq 2|V|$ apart, level repulsion

J. von Neumann, E. Wigner, Phys. Z. 30, 467 (1929)

Mean Level-Spacing Ratio





Gap ratio

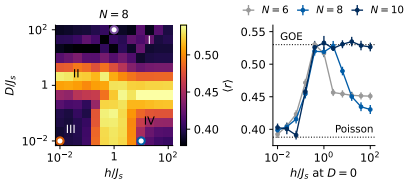
$r_n := \min(\delta_n, \delta_{n-1}) / \max(\delta_n, \delta_{n-1}) \in [0, 1]$, $\delta_n = E_{n+1} - E_n$: no unfolding

Poisson: $P(r) = 2/(1+r)^2$, $\langle r \rangle = 2 \ln 2 - 1 = 0.3863$; GOE: $\langle r \rangle \approx 0.530$

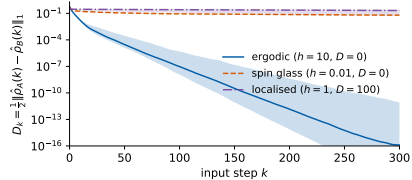
- chaotic register: few small gaps, histogram on the GOE curve; near-integrable: many small gaps, Poisson

V. Oganesyan, D. A. Huse, PRB 75, 155111 (2007); Y. Y. Atas *et al.*, PRL 110, 084101 (2013)

Chaos and Forgetting in Three Regimes



mean gap ratio $\langle r \rangle$ over field h and disorder D; Poisson 0.3863, GOE 0.530; circles: the three regimes; right panel: $D = 0$

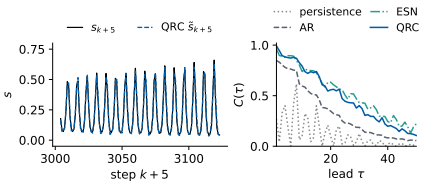


$\mathcal{D}_k = \frac{1}{2}\|\hat{\rho}_A(k) - \hat{\rho}_B(k)\|_1$, copies A, B with different starts, same random inputs; $\|\cdot\|_1$: sum of moduli of eigenvalues; median, band 16th to 84th percentile over 10 draws

- **ergodic** (strong field): nothing local conserved, GOE; start forgotten exponentially
- **spin glass** (weak field): $\hat{X}_j$ nearly conserved, Poisson; start kept
- **localised** (strong disorder): $\hat{Z}_i$ nearly conserved, Poisson; start kept
- $\langle r \rangle$ separates ergodic from non-ergodic; the other two differ in what is conserved

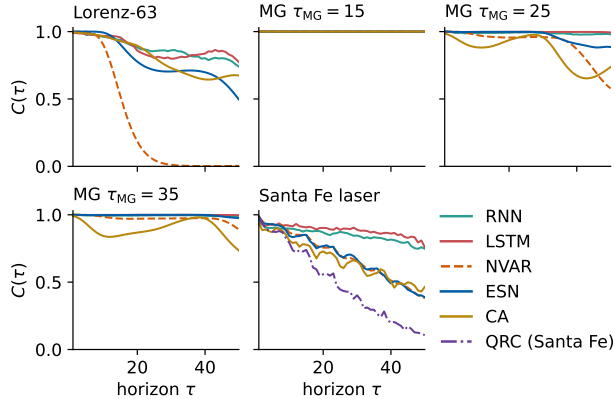
R. Martínez-Peña *et al.*, PRL 127, 100502 (2021)

Six Qubits on the Laser, Six Models Compared



QRC on Santa Fe: target s_{k+5} and forecast; $C(\tau)$ against τ; one draw of the couplings

- six qubits: well above AR and persistence, below an ESN with as many features



test $C(\tau)$ of RNN, LSTM, NVAR, ESN, CA and QRC (63 features, Santa Fe only)

Summary

reservoir	fixed state	forgets by	finding
ESN	leaky tanh units, random W	contracting random matrix	close to trained networks on Mackey–Glass
cellular automaton	bits, one local rule	overwriting input cells	moving rules with products, close to the ESN
quantum	six interacting qubits	discarding the input qubit	ergodic regime works; no advantage at equal size

- common: forgets its start and old inputs, responds nonlinearly; only a linear readout trained
- a reservoir that does not forget (spreading CA rules, near-integrable qubits) cannot be trusted by its readout

Thank you for your attention